Zero Trust-Based Security Implementation in ANPR System

Session 2022-2026

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1 Abstract

This project presents the development of a secured Automated Number Plate Recognition (ANPR) system using Zero Trust Architecture principles. ANPR systems are increasingly used for traffic monitoring, toll collection, and vehicle identification. However, security concerns such as unauthorized access, manipulated input data, and unverified user actions pose serious threats. To mitigate these, we integrated email verification, multi-factor authentication (MFA) using OTP via Email.js, and restricted access using Firebase authentication mechanisms. Our system only allows authenticated and verified users to access the dashboard, upload images, and invoke the machine learning model. Images are scanned to detect number plates, followed by Optical Character Recognition (OCR), and the resulting text is securely stored in Firebase. This report outlines the motivations, system design, Zero Trust integration, testing procedures, and outcomes, providing a secure framework that can be adopted in AI-driven applications.

2 Intoduction

The automated system of Number Plate Recognition (ANPR) identifies and analyzes vehicle license plates through artificial intelligence applications that process images. Modern organizations employ ANPR systems to perform traffic enforcement duties and manage entry points as well as fight crime incidents. The utility benefits of ANPR systems stand clear yet these systems introduce substantial security risks to the system. The key cybersecurity risks emerge from unauthorized network entries alongside altered picture inputs and API exposure points which both endanger vital data and permit improper AI functionality use.

Current perimeter security systems prove insufficient when confronting contemporary security threats. Under the Zero Trust Security model all requests must be verified before being trusted at any time. Zero Trust implements stringent identity verification while limiting access privileges and maintaining constant system monitoring because it treats every access request as if it were malicious.

The project team developed a secure ANPR system which complied with the fundamental principles of Zero Trust security architecture. Authentications run multiple times before allowing access and secure model access while maintaining data integrity through controlled authentication procedures. This paper examines both the design process and implementation and evaluation phase of the system.

3 Problem Statement

ANPR systems which operate through artificial intelligence face multiple security risks because of their designed nature.

- Unauthorised Access:Attackers gain unapproved entry to break into the system through unverified authentication processes.
- Manipulated Input: The model can fail to interpret false images designed by attackers.
- Data Breaches: Plate numbers alongside other sensitive information become vulnerable because improper access control measures fail to protect them.

The system resolves security concerns through Zero Trust implementation for user authentication and input validation with controlled access authorization.

4 Objectives

Objectives:

- The system must establish a protected user registration system that uses email confirmation protocols.
- The system requires users to set up MFA through OTP authentication managed by Email.js when they attempt to log in.
- Users must verify to gain access to all sensitive functionalities including dashboard access model invocation and data access
- The system requires image validation to validate their quality before upload.

- An AI model with OCR functionality accurately extracts license plate numbers from images.
- The system stores all generated results securely within Firebase databases using user-based permission controls and encryption.

5 System Architecture

5.1 Components

- Firebase Authentication: User registration, login, and verification
- Email.js: Sends OTP to user's email for second-factor authentication
- Dashboard: Frontend interface accessible only post-verification
- The Image Upload Module performs two functions: it validates image structure while ensuring correct image type before the processing step begins.
- Number plate detection and reading functions through machine learning form part of the AI Model Integration framework.
- Users can securely store text data through the Firebase Realtime Database under unique user IDs.

5.2 Workflow Overview

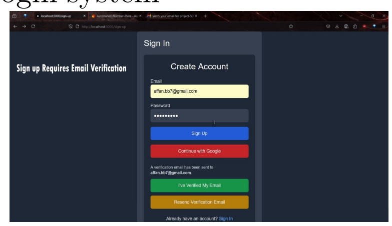
- The system generates a verification link that the user needs to validate through email.
- Once verification of the email completes users can log in to receive an OTP.
- The user gets dashboard access after validating their OTP.
- The system enables users to upload images that the platform validates.
- The AI model receives valid images for performing number plate detection.

- A script performs optical character recognition to extract detected plate numbers from the images.
- The application saves data in Firebase which users can view through the interface.

6 Zero Trust Implementation

6.1 Identity Verification

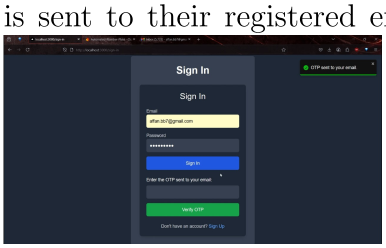
Users can only access the system through Firebase Auth after verification. After sign-up the system sends verification emails to users. The login system becomes available to users after they complete verification checks.



6.2 Two-Factor Authentication (2FA)

User authentication security received an improvement through the application implementation of two-factor authentication (2FA). The authentication method brings together two authentication factors where users bring their password knowledge and have access to their registered email to block unauthorized access.

- **U**sers must start login by entering their password as the first step. When users submit the correct password during the authentication process the system generates a one-time password (OTP) which they must receive through their registered email. Users must enter both their valid password along with their correct one-time password which is sent to their registered email address before accessing the system.



Account protection increases significantly because the 2FA approach reduces exposure to password reuse and phishing along with credential theft dangers.

6.3 Least Privilege Access

The protected routes dashboard, model and gallery are accessible only through proper authentication. Unauthenticated users cannot access the database because Firebase rules enforce this restriction.

6.4 Secure Model Access

Number plate recognition model operations can only be initiated by verified system users. The API blocks access for both unverified users and those whose session has expired.

6.5 Secure Communication (HTTPS)

Data security depended on the use of HTTPS which maintained secure communication across all client-server data transmissions. All communication between users and servers passes through the TLS (Transport Layer Security) encryption system which defends against spying activities and two attacks and data modification threats.

During development, testing the application made use of a self-signed SSL certificate to establish HTTPS functionality. The encryption provided by self-signed certificates enables secure communication verification but browser systems do not accept them for production-grade operations.

The configuration protects both client-server information exchange along with their data integrity throughout development.

6.6 Protection Against SQL Injection

Cloud Firestore protects the application from SQL injection attacks because it operates as a document-based NoSQL database. Firestore operates as a database distinct from SQL yet it does not accept raw query strings involving user input so dangerous SQL code injection remains impossible.

Data storage and modification through Firestore takes place through built-in method protocols and parameterized sequences that prevent user-made code execution. By removing traditional SQL queries made from user input the design reduces exposed weaknesses.

The application achieves default database security through Firestore without repetitive query sanitization.

6.7 Verification of images

6.7.1 Image Verification

To ensure the integrity and authenticity of uploaded image files, the following verification pipeline was implemented:

1. **Extension Validation:** The file extension was checked to confirm it belongs to a set of allowed image formats, such as `.jpg`, `.jpeg`, `.webp`, and `.png`. This acts as a preliminary filter against unsupported or malicious file types.

2. **Magic Number Inspection:** The first few bytes (magic number) of each file were read and compared against known image file signatures. For example, PNG files begin with `0x89504E47`, while JPEG files start with `0xFFD8`. This step prevents files with spoofed extensions from bypassing the system.

3. **Content Validation via Pillow:** The image is then opened using the Python Pillow library. The `Image.open()` and `.verify()` methods were used to parse and validate the internal structure of the image. This step detects corrupted or incomplete image data that might not be caught by earlier checks.

7 Results and Evaluations

An AI-powered ANPR platform used Zero Trust principles to achieve system success. Email verification combined with OTP-based MFA served as the identity validation mechanism. The application together with the database became inaccessible to unauthorized users through the implementation of strict Firebase rules. The AI model operated correctly after receiving inputs from validated workflows. The security measures proved effective because all bypass attempts failed to succeed.

8 Conclusion

The project proved how Zero Trust Architecture works effectively with AI-based systems and its practical implementation value. The system verified identities throughout all processes while restricting access points and implementing monitoring controls to prevent unauthorized activities and adversarial attacks. The system provides scalable real-world ANPR deployment security and its architecture extends to other high-security AI-based systems.